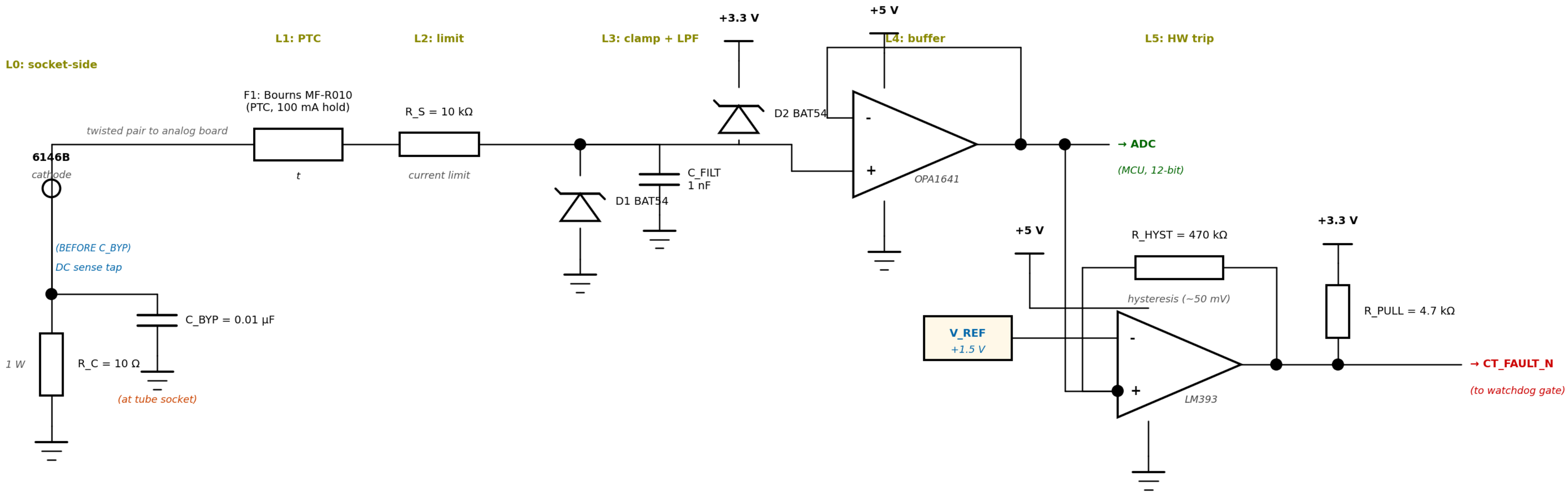
Per-Tube Cathode Current Monitor + Failsafe Chain

Seven defensive layers from cathode (+1 V nominal, may see 600 V on fault) to MCU ADC. Replicate one channel per tube; comparator outputs OR-ed to watchdog gate.



Operating point (per tube at full drive):	*I_cathode ≈ 100 mA → V_sense ≈ 1.0 V at ADC input → ~30% ADC range*
Trip threshold:	*V_REF = 1.5 V on LM393 (-) corresponds to 150 mA cathode current; 50 mV hysteresis from R_HYST.*
Total fault response (cathode overcurrent → tubes off):	*< 100 µs total -- op-amp slew (~1 µs) + comparator (~3 µs) + SR latch + screen-voltage gate discharge*
Critical layout rule:	*DC sense tap leaves R_C top BEFORE C_BYP. Sense wire is twisted pair (signal + analog GND return).*
Three separate grounds, one star point:	*GND_RF (at tube socket, bypass returns) | GND_SENSE (analog board signal GND) | GND_DIG (MCU board). All meet at the analog board single-point ground only.*

Figure 1. Per-tube cathode current monitor + failsafe chain. Levels 0-5 shown; Levels 6-7 (firmware) are described on the next page.

Defensive Layers and Bill of Materials

Level	Component(s)	Response time	Failure mode caught
L0	R_C (10 Ω, 1 W), C_BYP (0.01 µF NP0)	—	Normal sense + RF rejection at tube socket
L1	F1 = Bourns MF-R010 PTC (100 mA hold)	~ 1 s	Sustained fault thermal cooking of downstream parts
L2	R_S = 10 kΩ 1/4 W series limiter	instant	Fault current limited to 60 mA at 600 V cathode short
L3	D1, D2 = BAT54 Schottky clamps; C_FILT = 1 nF	instant	Voltage spikes beyond ±supply rails; RF anti-alias
L4	U1 = OPA1641 unity-gain buffer (+5 V / GND)	< 1 µs	ADC physically isolated; OPA1641 has ±20 V input protection
L5	U2 = LM393 comparator, V_REF = +1.5 V, R_HYST = 470 kΩ	≤ 100 µs	Hardware overcurrent trip → watchdog → screen-V drop
L6	Firmware ADC sampling at 1 kHz, threshold checks	~ 5 ms	Soft trip + warnings + UI alerts
L7	NVS fault log (timestamp, tube_id, type, current, duration)	persistent	Post-mortem analysis across power cycles

Bill of Materials (per tube; ×2 for the push-pair)

Ref	Part	Qty/tube	Notes
R_C	10 Ω 1 % 1 W metal film	1	Mount AT the tube socket. Vishay PR01000101000JR500 or similar.
C_BYP	0.01 µF NP0 ceramic, 100 V	1	Low-ESL package, ≤5 mm leads, at socket pin
F1	Bourns MF-R010 PTC	1	100 mA hold, 200 mA trip, V_max 60 V
R_S	10 kΩ 1 % 1/4 W metal film	1	Fault current limit
D1, D2	BAT54 (or 1N5817, PMEG3050)	2	Schottky clamps; consider PMEG3050 for fault-current margin
C_FILT	1 nF X7R 50 V	1	Anti-alias + RF reject
U1	OPA1641 single op-amp	1	Or OPA1642 dual for both tubes in one IC. Built-in ±20 V input protection.
U2	LM393 dual comparator	0.5	One IC handles both tubes
V_REF	LM4040DIZ-1.2 + 1 k trim	0.5 (shared)	Shared between both tubes' comparators
R_HYST	470 kΩ 1 %	1	Sets ~50 mV hysteresis
R_PULL	4.7 kΩ 1 %	1	Pull-up for LM393 open-collector output
C_DEC	100 nF X7R	4	Decoupling at each IC supply pin

Operating Points and Trip Levels

Condition	I_cathode	V_sense	ADC count (12-bit)	Action
Tubes cut off (key-up)	0 mA	0 V	0	normal idle
Nominal operating	100 mA	1.0 V	~1240	normal log
Soft warning threshold	120 mA	1.2 V	~1490	Firmware: warn UI, reduce envelope by 10 %
Hard fault threshold	150 mA	1.5 V	~1860	Hardware comparator trips → watchdog → screen drops to 0 V
Sustained overcurrent	200 mA+	2.0 V+	~2480+	PTC F1 also opens within ~1 s

Design Rationale and Procedures

Why screen-voltage interrupter (not plate, grid, filament)?

Plate: 600 V / 250 mA. Hard to switch reliably; relays slow, MOSFETs expensive at that V_{DS}. Capacitor discharge dumps significant energy.

Grid bias: already the firmware's normal off mechanism. Hardware fail-safe should be independent of firmware.

Filament: cathode thermal mass means seconds to cool. No protection against an in-progress fault.

Screen: low current (~30 mA), modest voltage (200 V), tiny supply cap → µs discharge. Without screen voltage, tubes are guaranteed off regardless of plate, grid, or filament state. Cleanest interrupter.

Calibration Procedure

1. Offset zero: Tubes in deep cutoff (key-up). Read each tube's ADC count; store as cathode_offset[tube_id].

2. Gain calibration: Insert a known shunt resistor in series with the cathode return, with a DMM measuring I_cathode. Drive the PA to multiple current levels (50, 75, 100, 125 mA). Record ADC count at each. Fit a line: I_cathode_mA = (ADC_count − offset) × gain. Store gain[tube_id].

3. Comparator threshold trim: Drive the tube to I_cathode = 150 mA. Adjust LM4040 divider trim until LM393 just trips. Verify it un-trips at 100 mA (hysteresis check). Store in NVS as trip_calibration_date.

4. Verification: Trip-test with the actual screen-voltage gate wired. Verify tubes go dark within 100 µs of the trip event. Use a scope on the screen pin to time-resolve the response.

Re-run calibration every 100 operating hours or after any servicing.

Layout and Grounding Rules

• **R_C and C_BYP mount AT the tube socket cathode pin**, with leads ≤ 5 mm. C_BYP's ground end goes to local chassis (GND_RF).

• **DC sense tap leaves R_C top BEFORE C_BYP.** If reversed, 14 MHz RF currents flow through the sense path instead of through C_BYP — sense becomes useless.

• **Sense wire is twisted pair**: hot signal + analog GND return. Common-mode pickup ≈ 0. Optionally shielded with shield grounded at the analog-board end only.

• **Three separate "grounds":** GND_RF (tube socket area, bypass caps); GND_SENSE (analog board signal); GND_DIG (MCU board). All three converge at exactly one star point on the analog board. No daisy-chain.

• **Op-amp and comparator on analog board**, NOT at the tube socket. Keeps them within their thermal envelope and away from the RF area.

• **+3.3 V rail to which D2 clamps**: needs ≥ 10 µF bulk cap near the clamp node to absorb fault energy without sagging.

Open Items

ADC choice: built-in ESP32-S3 ADC (12-bit, fast, ~80 µA LSB at 100 mA range) vs. ADS1115 external (16-bit, slow but precise via I^2C). Built-in is probably adequate.

SR latch: hardware (74HC74 or NAND latch from a second LM393 section) vs. pure firmware via esp_task_wdt GPIO. Hardware is < 100 µs, firmware ~1 ms. Hardware preferred for fault response speed.

BAT54 voltage rating margin: at 60 mA fault current sustained for ~1 s before PTC trips, BAT54 is at its specified peak. Consider PMEG3050 (higher current rating) or BAS70-04W as alternatives.

Per-pair sum monitor: optional third comparator with threshold at 280 mA total cathode current (sum of both tubes), to catch the case where both tubes drift hot together without individual tube exceeding its threshold.